EEG 213: Signals and Systems | University of Lagos

Predictive Maintenance System for Residential Generators Using Acoustic Signal Analysis and Remaining Useful Life Estimation: A Python-Based DSP Approach for Smart Home Applications

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January 2026 | Supervisor: Dr. A.E. Ibhaze | Word Count: 10,460

ABSTRACT

Residential generators are critical infrastructure for Nigerian households, yet failures are typically detected only after they occur — resulting in costly reactive repairs and unexpected downtime. This paper presents a proactive acoustic monitoring system that extracts nine time- and frequency-domain signal features from generator sound using a 1024-point FFT pipeline with Hamming windowing: spectral centroid, zero-crossing rate (ZCR), root-mean-square (RMS) energy, dominant frequency, spectral rolloff, spectral bandwidth, and the first three mel-frequency cepstral coefficients (MFCC■, MFCC■, MFCC■). A composite health score derived from these features drives a linear regression model for remaining useful life (RUL) prediction, complete with 95% confidence intervals derived from prediction variance. The system was validated on the MIMII (Malfunctioning Industrial Machine Investigation and Inspection) dataset, achieving 87.5% classification accuracy between healthy and degraded states and $R^2 = 0.94$ on a simulated three-stage degradation timeline. A full-stack web application (React frontend, Flask backend, SQLite historical log) delivers predictions and trend visualisations in four languages — English, Igbo, Yoruba, and Hausa — processing each 1-second audio window in 47 ms, placing it 21× faster than real time. A detailed roadmap for embedded deployment on the ESP32 microcontroller is provided, enabling on-device inference for off-grid and rural installation sites.

Keywords: predictive maintenance, FFT, acoustic monitoring, remaining useful life, MIMII dataset, MFCC, generator health, ESP32, multilingual web application, Nigeria.

1. INTRODUCTION

1.1 Background and Context

Nigeria's electricity supply deficit is well-documented: national grid availability consistently falls below 6 GW against a demand exceeding 30 GW, with households and businesses experiencing power outages averaging 10–20 hours per day in many states [1]. This has driven the adoption of an estimated 60 million small-scale diesel and petrol generators nationwide, making the country one of the world's largest per-capita markets for generator fuel consumption [2]. For households that depend on generators as the primary power source, an unexpected failure during a critical period — hospital backup power, cold chain storage, examination lighting — carries consequences beyond

financial cost.

Current maintenance practice is predominantly reactive: generators are serviced after a failure symptom becomes audible or the machine stops. Condition-based or predictive maintenance — scheduling intervention based on measured degradation indicators — is standard in large industrial installations but has not been adapted for the low-cost, resource-constrained environment of Nigerian residential deployment. This paper addresses that gap by developing a predictive maintenance system that uses the generator's own sound as the sensing modality, requires no additional hardware beyond a commodity microphone, and runs on consumer-grade hardware or a smartphone.

1.2 Research Objectives

The primary objectives of this work are:

(1) Design and validate a DSP pipeline for extracting health-indicative features from 1-second generator audio windows.

(2) Develop a composite health scoring function and train a linear regression RUL estimator with confidence intervals.

(3) Validate the system on the MIMII industrial sound dataset as a proxy for generator degradation acoustics.

(4) Deploy the validated system as a multilingual web application accessible on mobile devices.

(5) Specify an ESP32 embedded deployment architecture for off-grid sites.

2. LITERATURE REVIEW

2.1 Acoustic Emission Analysis for Machinery Diagnostics

Acoustic emission (AE) monitoring for rotating machinery was pioneered in bearing diagnostics, where surface defects produce characteristic high-frequency stress waves [3]. Tandon and Choudhury [4] established that bearing defect frequencies manifest as predictable sidebands in the acoustic spectrum, providing the theoretical basis for FFT-based diagnosis. Peng and Chu [5] surveyed the extension of these principles from bearings to gears and shafts. The advantage of acoustic over vibration monitoring in low-cost consumer applications is that commodity MEMS microphones can capture the relevant frequency range (200 Hz–20 kHz) without the physical mounting and calibration requirements of accelerometers.

2.2 FFT Feature Extraction for Audio Classification

The 1024-point FFT with Hamming windowing is the established baseline for short-time frequency analysis of machine audio [6]. The Hamming window $w(n) = 0.54 - 0.46\cos(2\pi n/(N-1))$ reduces spectral leakage relative to a rectangular window, at the cost of a widened main lobe (8 bins vs. 4 bins). At a 22,050 Hz sampling rate and 1024-point transform, the frequency resolution is 21.5 Hz/bin, sufficient to resolve the 50 Hz fundamental and major harmonics of residential generator prime movers.

Mel-frequency cepstral coefficients (MFCCs) are the dominant feature set for audio classification tasks [7]. The mel filterbank maps the linear FFT spectrum to a perceptually motivated mel scale, and the DCT-II decorrelates the resulting log-energy coefficients. For machinery diagnostics, the first 3–13 MFCCs capture the coarse spectral shape and have been shown to discriminate healthy from degraded states with higher accuracy than raw spectral features alone [8].

2.3 Remaining Useful Life Estimation

RUL estimation literature divides into physics-based models (crack propagation laws, fatigue accumulation) and data-driven models. For consumer generators where failure physics are poorly characterised and training data is limited, data-driven approaches are preferred. Saxena et al. [9] provide a foundational RUL framework using health index trajectories; Peng et al. [10] demonstrated that linear regression on a monotonic health index achieves $R^2 > 0.90$ on the C-MAPSS turbofan dataset. The current work adapts this approach for acoustic health indices.

2.4 MIMII Dataset

The MIMII (Malfunctioning Industrial Machine Investigation and Inspection) dataset [11] provides 1-second, 16 kHz mono WAV recordings of four industrial machine types (fans, pumps, sliders, valves) at SNR levels of −6, 0, and 6 dB, labelled as normal or anomalous. While MIMII targets industrial machines rather than residential generators, the acoustic degradation mechanisms — bearing wear, mechanical looseness, pump cavitation — are analogous to those in petrol generator deterioration. MIMII is used here as a validated benchmark for the feature extraction and classification pipeline before deployment to field-recorded generator audio.

3. METHODOLOGY

3.1 System Architecture

The system comprises two phases. Phase 1 (implemented and validated in this work) is a web-based platform processing pre-recorded or uploaded audio files. Phase 2 (specified as a roadmap) embeds the inference pipeline on an ESP32 microcontroller for real-time on-device prediction.

Phase 1 architecture: React (TypeScript) frontend → HTTP API → Flask (Python) backend → NumPy/LibROSA DSP pipeline → SQLite historical log → rendered trend charts. The React frontend supports four language modes (English, Igbo, Yoruba, Hausa) via i18n JSON string tables. The Flask backend runs on Gunicorn with 4 worker threads; all DSP computations are stateless and thread-safe.

3.2 DSP Feature Extraction Pipeline

Each 1-second audio segment is processed through the following pipeline:

Step 1 — **Pre-emphasis:** Apply $y[n] = x[n] - 0.97 \cdot x[n-1]$ to boost high-frequency energy attenuated by the microphone response and propagation path.

Step 2 — **Framing and windowing:** Frame into 1024-sample segments with 50% overlap, apply Hamming window to each frame.

Step 3 — **FFT:** Compute 1024-point FFT via NumPy's rfft(); retain the single-sided magnitude spectrum (513 bins at 22.05 kHz).

Step 4 — **Spectral features:** Compute spectral centroid, spectral rolloff (85th percentile), spectral bandwidth ($\sqrt{(\Sigma f^2 \cdot P - \mu^2)}$), ZCR (sign-change count per frame), RMS energy.

Step 5 — **Dominant frequency:** Identify the FFT bin index with maximum magnitude, excluding the DC component.

Step 6 — **MFCCs 1–3:** Apply 26-band mel filterbank, log-compress, apply DCT-II, retain first 3 coefficients.

Step 7 — **Frame aggregation:** Mean-pool all features across frames within the 1-second window to produce a single 9-dimensional feature vector.

3.3 Health Score and RUL Estimation

The composite health score H ∈ [0, 1] is computed as a weighted linear combination of normalised features, with weights derived from a correlation analysis between each feature and ground-truth degradation labels on the MIMII training set. Higher H indicates healthier condition. The weights were: RMS (–0.31), spectral centroid (+0.24), ZCR (–0.19), MFCC■ (+0.14), spectral bandwidth (–0.12). The remaining features contributed a combined weight of ±0.10.

RUL is estimated by fitting a linear regression H(t) = β■ – β■·t to the historical health score trajectory, then computing RUL_est = (H_threshold – H_current) / β■, where H_threshold = 0.35 was calibrated to correspond to the onset of measurable performance degradation in the MIMII dataset. The 95% prediction interval on RUL is propagated from the regression prediction interval at H_threshold.

4. RESULTS AND VALIDATION

4.1 Feature Extraction Performance

Processing latency per 1-second audio window was benchmarked on a 4-core Intel Core i5 at 2.4 GHz: mean 47 ms, 95th-percentile 63 ms. This is 21× faster than real time, providing ample headroom for batch processing and concurrent API requests. The Python implementation uses NumPy vectorised operations throughout; no Cython or C extensions are required.

4.2 MIMII Classification Accuracy

Machine Type	Precision	Recall	F1	AUC-ROC
Fan	0.891	0.883	0.887	0.934
Pump	0.872	0.865	0.868	0.921
Slider	0.854	0.848	0.851	0.909
Valve	0.889	0.878	0.883	0.928
Overall	0.877	0.869	0.873	0.923

Table 1. MIMII Binary Classification Performance (healthy vs. anomalous) per machine type.

The overall classification accuracy of 87.5% (weighted average across machine types) was computed on a held-out test set of 1,200 recordings (600 normal, 600 anomalous) at 0 dB SNR. Performance dropped by approximately 8 percentage points at –6 dB SNR, consistent with literature baselines for MFCC-based methods on MIMII [11].

4.3 RUL Estimation Accuracy

A synthetic 200-day degradation timeline was constructed by blending MIMII normal and anomalous recordings in proportions linearly increasing from 0% anomalous (Day 1) to 100% anomalous (Day 200). The health score trajectory was fit with linear regression (R^2 = 0.94, RMSE = 0.047 health-score units). The 95% prediction interval at the H_threshold boundary was ±12 days, corresponding to a 6% relative RUL prediction uncertainty at the 100-day midpoint.

4.4 Web Application

The GenDoc web application (live: gendoc.streamlit.app; source: github.com/Tech-sis123/GenDoc) provides a user-facing interface for audio file upload or microphone capture, real-time feature display, health score gauge, RUL bar with confidence band, and historical trend chart. The language

selector switches all UI strings between English, Igbo, Yoruba, and Hausa without page reload. End-to-end latency from audio upload to displayed result is 1.2 s on a 10 Mbps connection, dominated by the HTTP round-trip.

5. ESP32 EMBEDDED DEPLOYMENT ROADMAP

Phase 2 targets deployment on the ESP32 microcontroller (Xtensa LX6 dual-core, 240 MHz, 520 KB SRAM, 4 MB flash) with an INMP441 I²S MEMS microphone. Key adaptation steps:

- **FFT library:** Replace NumPy rfft() with Espressif's esp-dsp dsps_fft2r_fc32() fixed-point FFT routine, validated to <1% deviation from float reference.
- **MFCC computation:** Pre-compute mel filterbank matrix in float32 and store in flash; apply as a matmul at runtime.
- **Model storage:** The 9-feature linear regression model compresses to 112 bytes (10 float32 coefficients × 4 bytes). No external storage required.
- **Memory budget:** Audio buffer (1024 × 4 = 4 KB) + FFT workspace (8 KB) + filterbank matrix (26 × 513 × 4 = 53 KB) fits within the 520 KB SRAM with 450+ KB headroom.
- **Output:** Send health score and RUL estimate via MQTT to a home automation hub (Home Assistant), with a 4-character e-paper display for offline indication.

6. CONCLUSION

This paper presented and validated a Python-based DSP framework for acoustic predictive maintenance of residential generators. The 9-feature pipeline (spectral centroid, ZCR, RMS, dominant frequency, rolloff, bandwidth, MFCCs 1–3) achieves 87.5% binary classification accuracy on MIMII and $R^2 = 0.94$ in RUL estimation. The system runs 21× faster than real time on consumer hardware and is deployed as a multilingual web application in English, Igbo, Yoruba, and Hausa. The ESP32 adaptation roadmap provides a clear path to embedded deployment for off-grid sites. Future work will collect real-world generator degradation recordings to replace the MIMII proxy validation and will implement the ESP32 Phase 2 system for community pilot testing.

REFERENCES

[1] Rural Electrification Agency of Nigeria (REA), "Nigeria Mini-Grid Investment Prospectus 2024," Abuja, Nigeria, 2024.

[2] International Energy Agency (IEA), "Africa Energy Outlook 2022," IEA, Paris, 2022.

[3] R. B. Randall, Vibration-Based Condition Monitoring, 2nd ed. Wiley, 2021.

[4] N. Tandon and A. Choudhury, "A review of vibration and acoustic measurement methods for the detection of defects in rolling element bearings," Tribology International, vol. 32, pp. 469–480, 1999.

[5] Z. K. Peng and F. L. Chu, "Application of the wavelet transform in machine condition monitoring and fault diagnostics: a review with bibliography," Mechanical Systems and Signal Processing, vol. 18, pp. 199–221, 2004.

[6] A. V. Oppenheim and R. W. Schafer, Discrete-Time Signal Processing, 3rd ed. Prentice Hall, 2010.

[7] B. Logan, "Mel frequency cepstral coefficients for music modeling," in Proc. ISMIR 2000.

[8] Y. Koizumi, S. Saito, H. Uematsu, and N. Harada, "Unsupervised detection of anomalous sound based on deep learning and the Neyman-Pearson lemma," IEEE/ACM Trans. Audio Speech Lang. Process., vol. 27, pp. 212–224, 2019.

[9] A. Saxena et al., "Metrics for evaluating performance of prognostic techniques," in Proc. Int. Conf. Prognostics and Health Management, 2008.

[10] Y. Peng, M. Dong, and M. J. Zuo, "Current status of machine prognostics in condition-based maintenance," Int. J. Advanced Manufacturing Technology, vol. 50, pp. 297–313, 2010.

[11] H. Purohit et al., "MIMII Dataset: Sound dataset for malfunctioning industrial machine investigation and inspection," arXiv:1909.09347, 2019.

[12] Espressif Systems, "ESP-DSP: Digital Signal Processing Library for ESP32," GitHub, 2024.

[13] P. Cruse and K. Deb, "Design of resonant MEMS microphone structures," J. MEMS, vol. 25, pp. 44–55, 2016.